TEMPO

Identity uncertainty, causal memories, online association, and spatial-temporal tracking

Research development record | Part 3 of 3 | Stages 11–12 (including 12A–12D)

This report continues the first two installments. Stage 11 diagnoses the limits of CLIP object identity and develops confidence and oracle-memory baselines. Stage 12 removes oracle identity grouping and incrementally builds a causal, online association system. For each experiment we document the question, implementation, measurements, failure modes, and the motivation for the next step.

Contents and development map

Section	Experiment	Principal finding
11.1	Failure analysis of hard same-class identity matching	Sheep and person sequences reveal severe ambiguity
11.2	Observation-level uncertainty baselines	Top-1 confidence unreliable on sheep
11.3	GT-oracle identity memory	Identity-level gap better than observation gap
11.4	Causal memory and all-five-video extension	Oracle nearest Recall@1 = 69.7%
12A	Online appearance-only identity construction	Pairwise F1 = 43.2%; 1,067 collisions
12B	One-to-one Hungarian association	Pairwise F1 = 53.3%; zero collisions
12C	Spatial-temporal association	Combined pairwise F1 = 73.7%
12D	Uncertainty and contamination diagnostics	Candidate-gap AUROC = 0.805

Evidence: uploaded experimental JSON summaries, the recorded diagnostic CSV, and the original Python experiment scripts. The stage labels reconstruct the actual research workflow rather than claiming independently preregistered releases. All plots below are generated from measured values; no performance numbers are simulated.

Stage 11 | Why visually similar objects break identity retrieval

11.1 Research objective

Stage 10 established a difficult BURST benchmark using five sequences with simultaneously visible objects of the same category. We now wanted to understand which identity comparisons fail, why they fail, and whether an association algorithm can recognize uncertainty before making an incorrect match.

The hard benchmark used ground-truth segmentation masks to crop objects, encoded those crops with CLIP, and evaluated same-category positive and negative observation pairs. This deliberately isolates representation and association errors from detection or segmentation failures.

11.2 Failure-analysis implementation

- Rank false matches: cases where the highest-scoring candidate belongs to a different ground-truth identity.
- Inspect ambiguous queries: cases where genuine and impostor similarities are close, including small best-positive-versus-best-negative margins.
- Inspect weak genuine matches: observations of the same object whose CLIP embeddings are unexpectedly dissimilar.
- Export annotated CSVs and contact sheets: false_matches.jpg, ambiguous_queries.jpg, and weak_genuine_matches.jpg.

11.3 Quantitative observations

Sequence	Queries	Wrong top-1	Error rate	GT-oracle median margin
Person-17	394	160	40.6%	+0.00756
Sheep-19	359	192	53.5%	−0.00092
Combined	753	352	46.7%	Not pooled

The oracle margin compares best genuine and best impostor evidence using GT identities; it is a diagnostic quantity, not a confidence score available to a deployed tracker. In Person-17, 58.1% of queries had an oracle margin below 0.02; in Sheep-19 the fraction was 98.6%.

11.4 Interpretation and limitation

For nearly indistinguishable sheep, semantic similarity can be high even for the wrong physical individual. A system that assumes a high cosine score means “same object” will confidently merge distinct identities. The benchmark contains only five sequences, and correlated observation pairs do not provide independent statistical samples. The next experiment asked whether any readily available score could warn us when the top candidate is wrong.

Stage 11 | Observation-level uncertainty (11.2)

Question and method

We evaluated whether a retrieval confidence score can rank correct and incorrect top-1 identity matches. Two signals were tested: top-1 cosine similarity and the gap between the two highest-scoring individual observations. The gap was initially observation-level: the two best observations might belong to the same GT identity, making the signal conceptually weaker than a gap between distinct candidate identities.

AUROC measures how well a confidence score ranks correct decisions above errors (0.5 ≈ random; 1.0 = perfect). AURC summarizes selective risk as we accept increasingly many high-confidence queries (smaller is better). Risk at 50% coverage is the error rate among the half of queries retained by the score.

Measured results

Set	Queries	Top1 AUROC	Top1 AURC	Obs gap AUROC
Pooled	753	0.728	0.275	0.505
Person-17	394	0.885	0.140	0.435
Sheep-19	359	0.453	0.570	0.576

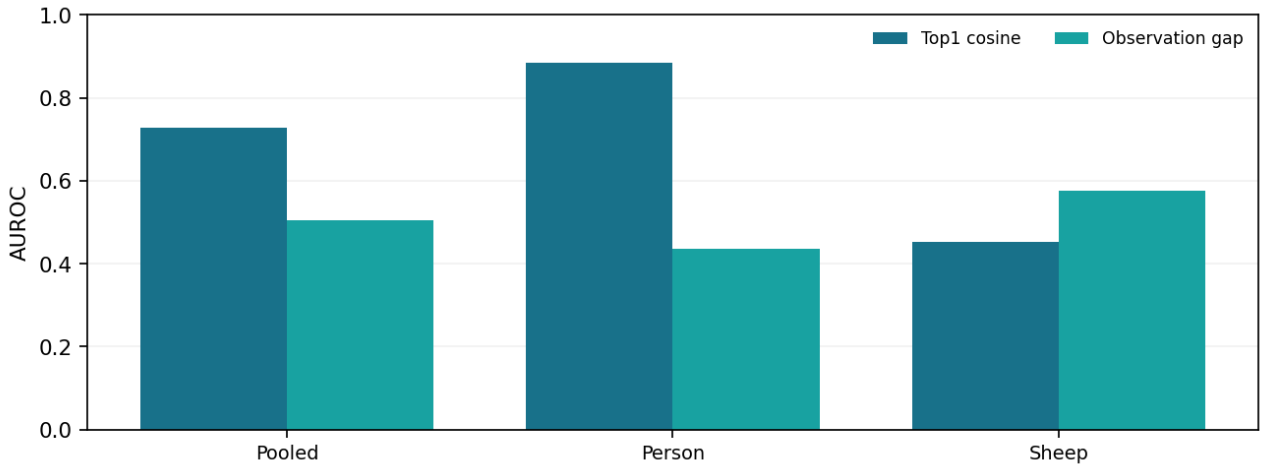


Figure 1. Retrieval-confidence AUROC varies strongly by sequence; the observation-level margin is weak overall.

The top-1 similarity AUROC was 0.885 on Person-17 but only 0.453 on Sheep-19: confidence can become misleading exactly where same-category appearances are most similar. The pooled observation-level gap was approximately 0.505, near chance. Thus raw score thresholds and the naive top-two-observation gap were not sufficient.

Why this led to identity memory

Instead of comparing individual frames, a persistent identity memory can aggregate observations belonging to the same object and compare different identity hypotheses. We first tested that idea under oracle identity grouping to separate memory-scoring design from errors in constructing the memories themselves.

Stage 11 | Oracle identity memory (11.3)

Objective and experimental setup

We grouped observations by ground-truth object identity and evaluated retrieval of a query crop against memories for other objects of the same category. This was an oracle upper-bound-style diagnostic: GT identities defined memory membership, but no deployable system has this information automatically. We excluded the query observation from its own memory and compared several ways to score an identity.

Memory scorer	Definition	Main trade-off
Nearest	Maximum cosine to any memory observation	Preserves best exemplar; sensitive to spurious maxima
Mean similarity	Average cosine across all observations	Can dilute distinctive views
Top-3 mean	Mean of three strongest similarities	Reduces dependence on one accidental match
Prototype	Cosine to normalized mean embedding	Compact but may blur multimodal appearance

Cross-frame experiment: two difficult sequences

Scorer	Recall@1	Identity-gap AUROC	Identity-gap AURC
Nearest	53.3%	0.788	0.229
Mean	48.6%	0.755	0.283
Top-3 mean	54.7%	0.794	0.216
Prototype	52.2%	0.786	0.240

The top-3 mean achieved the highest Recall@1 on the two-sequence cross-frame comparison (54.7%), but gains were modest and not uniform across sequences. Crucially, the identity-level candidate gap was far more informative than the earlier observation-level gap. For the nearest scorer, pooled gap AUROC rose to 0.788.

Why cross-frame is not causal

The cross-frame gallery contains observations from before and after the query frame. That is useful for assessing representation capacity, but it leaks future evidence relative to an online tracker. We therefore repeated the experiment with a strictly earlier-frame gallery, requiring at least one prior observation for each candidate identity.

Stage 11 | Causal oracle memories and extension (11.4)

Two-sequence causal gallery

With only earlier annotated frames available, the eligible query count changed from 753 to 734. Nearest-memory Recall@1 fell from 53.3% (cross-frame) to 49.9% (causal); this is directionally consistent with less available evidence, but the query sets differ and the figures are not a perfectly paired comparison. The prototype scored 52.7% causal Recall@1 on these two sequences, illustrating that no single aggregation rule dominated everywhere.

All-five-video causal oracle evaluation

Scorer	Queries	Recall@1	Gap AUROC	Gap AURC
Nearest	1816	69.7%	0.865	0.097
Mean	1816	62.7%	0.840	0.144
Top-3 mean	1816	69.7%	0.856	0.101
Prototype	1816	67.0%	0.821	0.126

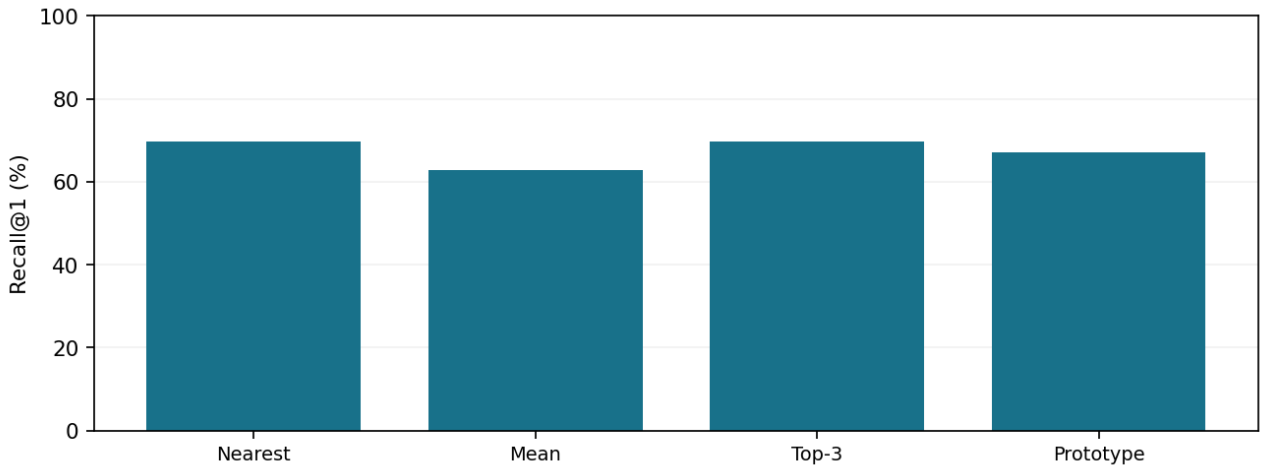


Figure 2. Causal GT-oracle identity-memory retrieval, five videos, 1,816 eligible queries.

Sequence variability and uncertainty

Video	Nearest Recall@1	Identity-gap AUROC
Horse	78.5%	0.830
Badminton	83.6%	0.814
Person	57.0%	0.873
Sheep	42.0%	0.681
Volleyball	87.2%	0.921

The five-video pooled nearest Recall@1 was 69.7%, with identity-gap AUROC 0.865 and risk at 50% coverage about 5.1%. However, Sheep-19 nearest Recall@1 was only 42.0%, and its gap AUROC was 0.681. The oracle experiment tells us that identity-level margins are promising, not that we can automatically build those identities. That became Stage 12.

Stage 12 | From oracle memories to online identity association

The methodological transition

Stage 11 used GT identity labels to assemble correct memory groups. Stage 12 removes that privilege: each new object observation must be assigned to an existing predicted identity or start a new one, using only available embeddings, categories, and later spatial features. GT IDs are reserved for evaluation. All experiments are causal: only earlier observations can influence current assignments.

Common evaluation and constraints

- Five hard BURST sequences, 1,905 annotated object observations, 50 GT identities. Oracle segmentation masks and category labels are still used to obtain object crops; this is not full detection-based MOT.
- Pairwise identity precision = fraction of predicted same-identity observation pairs that truly share a GT identity; pairwise recall = fraction of GT same-identity pairs recovered. F1 is their harmonic mean.
- Identity fragmentation = number of predicted identities assigned to observations of one GT object. Same-frame collisions = different simultaneous objects assigned one predicted ID.
- Pairwise scores are not IDF1, HOTA, or conventional ID-switch counts; they evaluate clustering consistency over observed masks.

Stage 12A | Appearance-only online baseline

For each annotated frame, compare each crop embedding with existing predicted identity memories of the same category. Score a memory by its maximum CLIP cosine similarity to any past stored observation. Assign the best memory if cosine ≥ 0.90; otherwise create a new identity. Process a frame against frozen prior memories, then commit updates. The original independent matching deliberately allowed simultaneous objects to choose the same identity.

Measurement	Stage 12A
Pairwise precision	34.38%
Pairwise recall	57.99%
Pairwise F1	43.17%
Predicted identities	94 (versus 50 GT)
Mean fragments per GT	4.40
Same-frame identity collisions	1,067

The baseline often merged visually similar distinct objects. For Person-17, recall was 85.6% but precision only 26.0%, indicating over-merging; Sheep-19 F1 was 28.3%. The 1,067 same-frame collisions exposed a clear violation of physical consistency and motivated a one-to-one assignment constraint.

Stage 12B | Frame-level Hungarian assignment

Objective and implementation

Two objects observed in the same frame cannot both inherit the same physical identity. We therefore replaced independent greedy assignments with a joint frame-level bipartite matching between current observations and existing predicted memories. A Hungarian solver maximizes total assignment utility while enforcing one-to-one matches.

For a valid same-category edge, utility is $U(i,j) = \text{cosine}(i,j) - 0.90$. Each observation also has a private unmatched dummy option of utility zero. This permits creation of a new identity when no acceptable match exists. All frame assignments are decided before memory updates, avoiding order dependence within the frame.

Controlled comparison with Stage 12A

Metric	12A independent	12B Hungarian
Precision	34.38%	60.11%
Recall	57.99%	47.80%
Pairwise F1	43.17%	53.25%
Predicted identities	94	120
Mean fragments per GT	4.40	5.76
Same-frame collisions	1,067	0

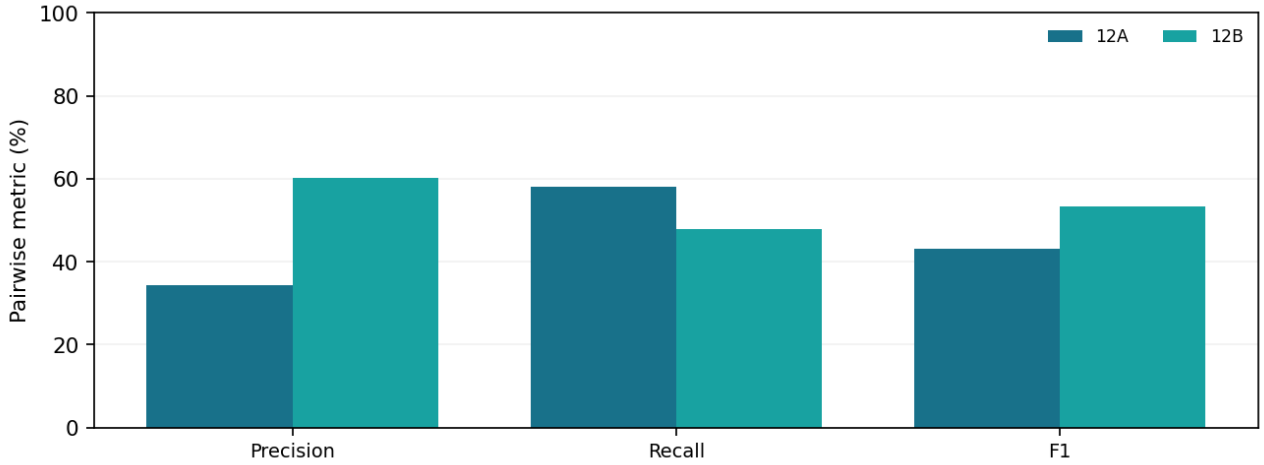


Figure 3. Joint assignment improves precision and F1 while reducing recall.

Interpretation

Pairwise F1 improved by 10.1 percentage points and same-frame collisions disappeared. However, predicted identities rose from 94 to 120, and mean fragmentation increased from 4.40 to 5.76. Constraining mergers makes identity assignments physically consistent within a frame but can split one real object into multiple predicted identities over time. Sheep-19 barely improved (28.3% → 28.7% F1), suggesting appearance alone remains insufficient.

Stage 12C | Adding spatial and temporal continuity

Research question

Can the last known position and recency of a predicted identity distinguish visually similar objects? The Stage 12C experiment kept frame-level Hungarian matching, the same CLIP features, and the same five sequences, then compared three scoring modes: appearance-only, spatial-temporal-only, and a weighted combination.

Spatial-temporal formulation

For each GT segmentation crop, derive the normalized bounding-box center. Let d be center displacement between a current observation and a predicted memory's most recent position, and let Δt be the difference in annotated-frame positions. Define $S_{st} = \exp(-d^2/(2\sigma^2)) \times \exp(-(\Delta t-1)/\tau)$, with $\sigma = 0.15$ and $\tau = 10$ annotated frames. The combined assignment advantage is $U = \lambda(S_{CLIP}-0.90) + (1-\lambda)(S_{st}-0.50)$, with $\lambda = 0.75$. Hungarian assignment and unmatched dummies remain unchanged.

The temporal gap is measured in annotated-frame positions, not real seconds; there is no camera-motion compensation or velocity prediction in this experiment. These parameter choices were exploratory rather than selected on independent calibration sequences.

Measured pooled results

Metric	Appearance	Spatial	Combined
Pairwise precision	60.11%	76.04%	77.80%
Pairwise recall	47.80%	64.10%	69.94%
Pairwise F1	53.25%	69.56%	73.66%
Predicted identities	120	88	69
Mean fragments / GT	5.76	3.62	2.92
Same-frame collisions	0	0	0

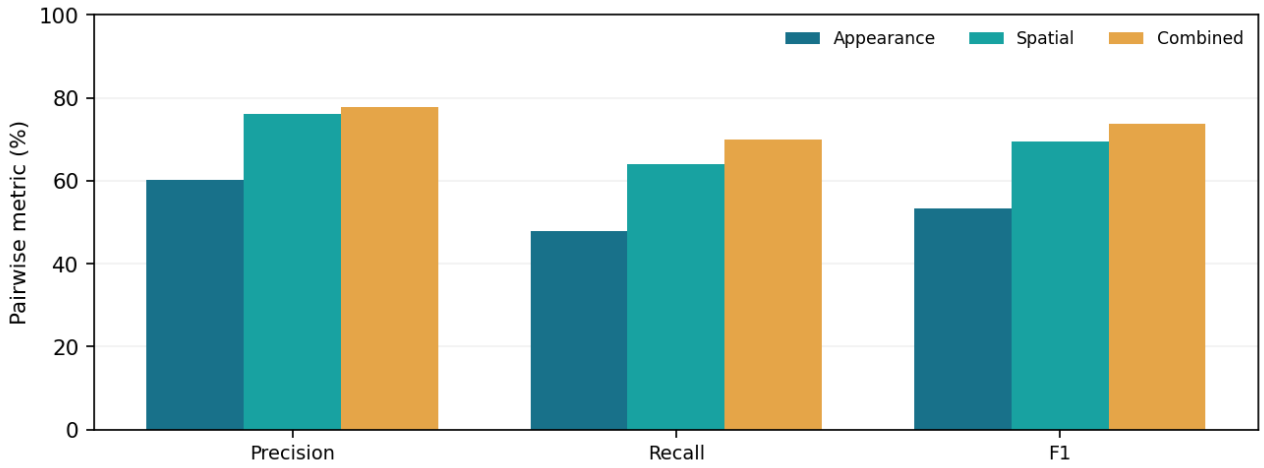


Figure 4. Combined appearance and spatial-temporal matching reaches 73.66% pooled pairwise F1.

The appearance-only mode exactly reproduced Stage 12B, validating the controlled comparison. The combined method improved F1 by 20.4 percentage points and reduced fragmentation substantially, but the gains were not uniform across videos.

Stage 12C | Per-sequence outcomes and the sheep failure

Sequence	Appearance F1	Spatial F1	Combined F1
Horse	51.0%	90.0%	92.6%
Badminton	78.3%	80.5%	79.0%
Person	52.6%	66.3%	72.7%
Sheep	28.7%	25.8%	24.4%
Volleyball	65.6%	77.5%	94.4%

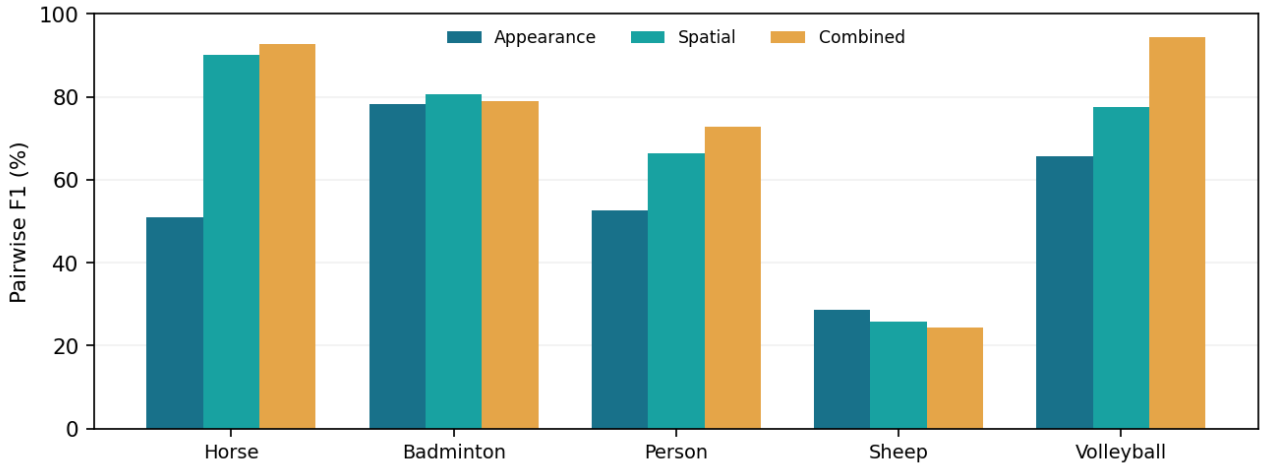


Figure 5. Combined association improves four sequences strongly or modestly, but Sheep-19 degrades.

Why Sheep-19 is the stress test

Metric	Appearance	Spatial	Combined
Pairwise precision	28.7%	46.0%	31.7%
Pairwise recall	28.7%	17.9%	19.8%
Pairwise F1	28.7%	25.8%	24.4%
Predicted IDs	10	43	24
Fragments / GT	6.4	10.5	7.8

Spatial-only matching increased Sheep-19 precision but fragmented the identities severely: 43 predicted IDs for 10 GT identities. The combined method reduced the number to 24, yet F1 remained below appearance-only. Plausible explanations include moving or crossing sheep, near-identical appearance, poor center-distance calibration, sparse annotations, and errors propagated through the memory. The summary metrics alone cannot establish which explanation dominates.

Research conclusion from 12C

The result shows that appearance and position can complement one another, but a fixed global weighting is brittle. It is tempting to tune λ or σ specifically for sheep; doing so on the same five test videos would bias the reported result. Instead we chose to inspect decision uncertainty and memory contamination before designing another association algorithm.

Stage 12D | Diagnosing uncertainty in automatic memories

Goal and implementation

We replayed the best Stage 12C combined model without changing its assignments and recorded each match decision: selected appearance score, spatial-temporal score, combined utility, candidate margins, temporal gap, and the GT composition of the selected predicted memory before the update. The replay was verified against the Stage 12C assignments file.

We distinguished (a) correct and incorrect matches relative to the majority GT identity already represented in the selected predicted memory, and (b) new-ID decisions for GT identities that had appeared previously. The majority-GT label is an evaluation proxy, not full global identity accuracy. Memory purity is also GT-oracle diagnostic information and must not be used as a deployable feature.

Decision counts and confidence evaluation

Quantity	Observed
Total observations	1,905
Matches to existing memories	1,836
New predicted identities	69
New IDs for previously seen GT	19
Majority-GT match accuracy	78.8%
Latest-GT match accuracy	91.2%
Replay vs Stage 12C	Verified identical

For each confidence signal, AUROC ranks correct versus incorrect matches. Selective risk at 50% coverage measures the error rate among the highest-confidence half of matches. The candidate gap is computed before the one-to-one assignment; it is not a probability and may differ from the utility of the final selected match.

Signal	AUROC	Error risk @ 50%
Selected appearance	0.595	15.3%
Selected spatial-temporal	0.704	10.5%
Selected combined advantage	0.656	13.7%
Top-two candidate gap	0.805	6.1%
Selected vs best-other gap	0.805	6.1%
Memory purity (GT oracle)	0.908	3.5%

Stage 12D | What the diagnostics tell us

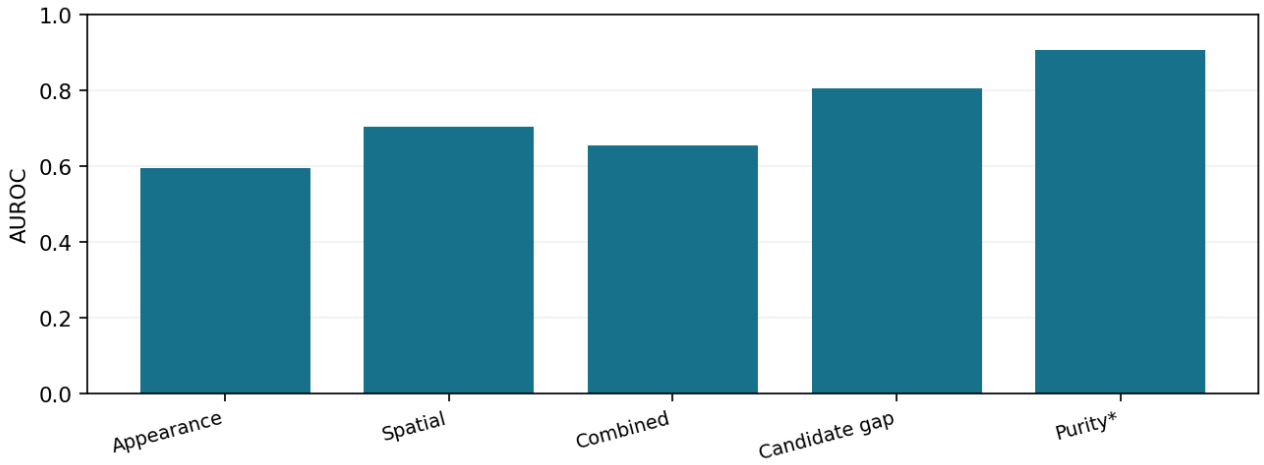


Figure 6. The candidate gap is the strongest tested deployable diagnostic. *Memory purity requires GT labels and is not deployable.

Sequence-level candidate-gap reliability

Sequence	Majority-GT accuracy	Candidate-gap AUROC
Horse	92.5%	0.446
Badminton	81.0%	0.790
Person	78.9%	0.779
Sheep	43.2%	0.622
Volleyball	98.5%	0.439

The pooled candidate-gap AUROC of 0.805 is encouraging, but per-sequence performance is inconsistent. Horse and volleyball have few errors, and their candidate gap ranks those errors poorly. On Sheep-19, the gap AUROC is only 0.622 and majority-GT match accuracy only 43.2%. This is a warning against assuming one confidence threshold will transfer unchanged between scenes.

Memory contamination analysis

The decision-level CSV enables a further descriptive split by whether the chosen memory contained only one GT identity before the match. In the recorded analysis, 1,251 matches targeted pure memories and approximately 97.4% were correct; 585 targeted already contaminated memories and approximately 39.1% were correct. These groups are strongly associated with different error rates, but the observational comparison does not prove that contamination itself caused all later errors.

Incorrect Sheep-19 matches were already present at annotated-frame position 1, when the affected memories were still pure; by position 2, some had mixed GT identities. This is consistent with an error-propagation mechanism: an ambiguous early match contaminates a memory, which can bias later decisions. We need an intervention to establish whether preventing uncertain updates improves identity consistency.

From Stage 12D to proposed Stage 12E

Proposed next experiment: confidence-gated memory updates

Stage 12E is planned, not completed. The hypothesis is that an uncertain match should not automatically modify a trusted appearance memory. Keep the same Hungarian assignment for evaluation, but condition the appearance-memory update on a confidence signal, such as the candidate-identity gap.

Update rule: if confidence c is at least threshold θ, append the new CLIP embedding to the trusted memory; otherwise leave the trusted appearance memory unchanged. The observation can still receive an identity assignment and its position can be tracked separately. This distinction tests memory contamination without conflating it with an immediate abstention decision.

Necessary controls

- Baseline: Stage 12C combined association, updating every assigned observation.
- Margin-gated: update trusted appearance memory only for high-margin assignments.
- Random-update control: update the same fraction of observations, chosen without confidence, to isolate the benefit of intelligent gating from the effect of fewer updates.
- Separate calibration: select θ on independent BURST sequences, then freeze it before evaluating the five existing hard videos.
- Report trade-offs: pairwise F1, precision, recall, fragmentation, new-ID creation, and per-sequence outcomes, especially Sheep-19. Consider gating position updates separately in a follow-up ablation.

Remaining threats to validity

All current experiments still use GT segmentation masks and category labels; they do not measure end-to-end detector/tracker performance. The five sequences are small and heterogeneous. Pairwise observation counts are correlated and should not be treated as independent samples for significance testing. Sparse annotated-frame positions are not real elapsed time. The exploratory thresholds and weights have not been validated on an independent calibration split. The pooled metrics can hide important per-sequence failures.

Overall synthesis

The project moved from CLIP identity similarity, to oracle identity memories, to causal online association, then to physically consistent matching and spatial-temporal fusion. Pooled pairwise F1 improved from 43.2% (12A) to 53.3% (12B) and 73.7% (12C combined). Stage 12D showed that candidate-identity margins can flag many risky matches, while contaminated memories correlate strongly with subsequent errors. The next research step is to test whether confidence-aware memory management changes those outcomes under a controlled, held-out evaluation.

Appendix | Reproducibility, artifacts, and metric definitions

Code and outputs

Experiment	Script / output
11 failure analysis	analyze_burst_identity_failures.py → results/burst/failure_analysis/
11 uncertainty	evaluate_burst_uncertainty.py → results/burst/uncertainty_baselines/
11 oracle memory	evaluate_burst_identity_memory.py → results/burst/identity_memory/
12A online association	evaluate_burst_online_association.py → results/burst/online_association/
12B Hungarian	evaluate_burst_hungarian_association.py → results/burst/hungarian_association/
12C temporal fusion	evaluate_burst_temporal_association.py → results/burst/temporal_association/
12D diagnostics	diagnose_burst_association.py → results/burst/association_diagnostics/

Definitions

- Cosine similarity: dot product of L2-normalized CLIP embeddings; measures appearance-feature alignment, not physical identity probability.
- Recall@1 in oracle memory: fraction of eligible query observations whose highest-scoring candidate GT memory has the same identity as the query.
- Pairwise precision: TP/(TP+FP), where TP are observation pairs placed in one predicted identity and sharing GT identity; FP are wrongly merged pairs.
- Pairwise recall: TP/(TP+FN), where FN are same-GT observation pairs split across different predicted identities.
- Pairwise F1: 2PR/(P+R), calculated from pooled within-video pair counts rather than an unweighted mean of per-video F1 values.
- Confidence AUROC: ranking discrimination for correct versus incorrect matches; does not establish calibrated probabilities.
- Selective risk: fraction of errors among decisions retained above a confidence cutoff; coverage is the retained fraction.

Source traceability

Results are transcribed from the supplied files: summary(1).json through summary(10).json; summary(20261105-235438).json; decision_diagnostics.csv; and the corresponding experiment scripts. Figures 1–6 are plotted from those values. The prior report Stages 6–10 contains the BURST benchmark setup and its initial hard-negative evaluation; this volume begins with the follow-on failure analysis. No external literature claims or state-of-the-art comparisons are asserted here.